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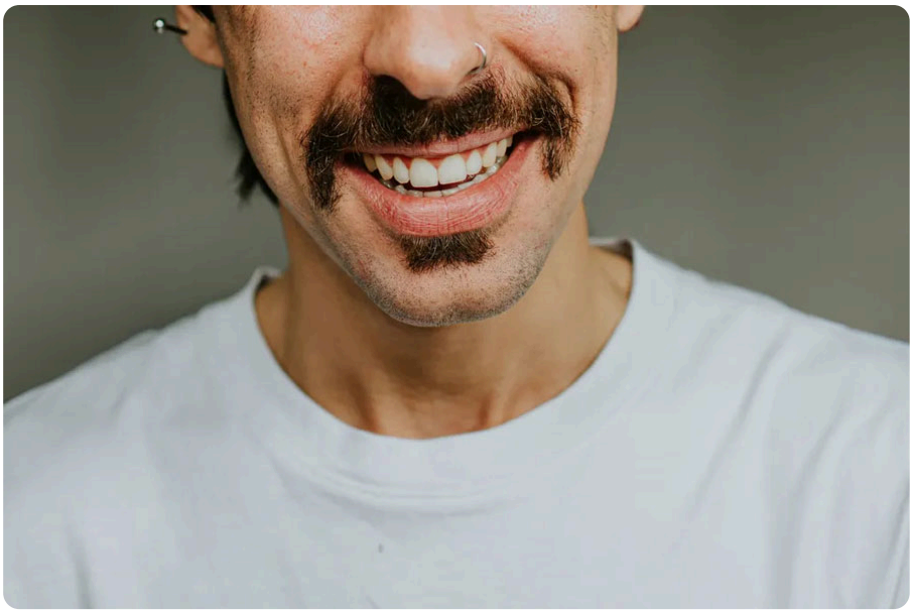

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Why are humans the only species with a chin?

Potential explanations abound, yet recent research has shed new light on the question.

Humans are the only species with a chin — a feature absent from even our closest relatives. Indeed, it's such a unique anatomical quirk that it's one of the main traits anthropologists use to identify Homo sapiens remains in the fossil record.

Yet, for such a defining feature, we know surprisingly little about its evolutionary purpose. So why are we the only species with a chin?

This question is hard to answer because experts haven't agreed on a single definition of a chin. While some researchers have argued that animals like elephants and manatees have chin-like protrusions, they're not the same T-shaped structures that protrude beyond our own bottom teeth. As a result, some

scientists have moved away from thinking of the chin as a single trait, instead referring to it as the collective result of interactions between many different parts of our head and jaw.

"So much about the chin is complicated," said Scott A. Williams, an evolutionary morphologist at New York University. "It cannot be quantified by a single metric but is rather composed of a constellation of morphological features."

A better understanding of the chin's function, in turn, could help scientists craft a definition. Experts have proposed several possible purposes for the chin.

Some have suggested that as we evolved smaller teeth, the chin appeared to reinforce our lower jaw and keep our teeth from breaking as we chewed. Others believe the chin may be linked to yet another unique human trait — our capacity for speech — with the chin providing an anchor point for our tongue muscles. And still others say the variation in how pronounced our chins are offers a hint that it could be linked to sexual selection.

Noreen von Cramon-Taubadel, an evolutionary morphologist at the University at Buffalo in New York, set out to winnow that list by determining whether the chin could have evolved by random chance or if evolution has been acting upon it directly.

To do so, von Cramon-Taubadel and her team studied dozens of traits linked to head and mandible size, including nine traits associated with the chin. Then, using an evolutionary tree of 15 hominoids — a group that includes humans, their fossil ancestors, gorillas, chimpanzees, orangutans and gibbons — they looked at whether those traits have changed more or less over time compared to random chance. Either result would suggest a role for natural selection in the evolution of the lower jaw.

Compared with other species, "the human cranium is more different from our ancestors' than we would expect given how much time has passed," she said. However, only three of the nine chin-specific traits appeared to be under direct selection.

Together, the team's results, published in the journal PLOS One, suggest the chin may be what's known as a spandrel — a term borrowed from architecture to describe a feature that is a side effect of something else. Coined by evolutionary biologists Stephen Jay Gould and Richard Lewontin in 1979, the

concept of a spandrel was introduced to argue against the view that every feature must serve a specific, evolved purpose.

"Instead, it appears that structurally, we have to have a chin, but not because the chin evolved to have a particular function," von Cramon-Taubadel told Live Science. "More and more studies are showing that things that we used to think were terribly important in terms of differences between humans and other apes actually could evolve just by random drift and gene flow."

Von Cramon-Taubadel said the group's findings appear to be more strongly influenced by known major landmarks in human evolution, including when we started walking upright and growing larger brains.

Despite these takeaways, von Cramon-Taubadel and Williams agree that the question is far from settled. It's unknown, for example, when traits like speech first appeared, so it's difficult to link them to chin evolution. While Williams accepts that the chin may not have evolved for a specific purpose, that doesn't make it arbitrary.

"It is still one of the defining features of our lineage that is present in some form in every human living on the planet today," he said.